

Circles

Problem 1. Let $ABCD$ be a cyclic quadrilateral. Let E be the intersection of lines parallel to AC and BD passing through points B and A , respectively. The lines EC and ED intersect the circumcircle of triangle AEB again at F and G , respectively. Prove that points C, D, F , and G lie on a circle.

Problem 2. Let $ABCD$ be a trapezoid with $AB \parallel CD$ inscribed in a circle Γ . Let P and Q be two points on segment AB (where A, P, Q, B appear in that order and are distinct) such that $AP = QB$. Let E and F be the second intersection points of lines CP and CQ with Γ , respectively. Lines AB and EF intersect at G . Prove that line DG is tangent to Γ .

Problem 3. Let P be the intersection of AC and BD in isosceles trapezoid $ABCD$ (with $AB \parallel CD$ and $BC = AD$). The circumcircle of triangle ABP intersects BC for the second time at X . Point Y lies on AX such that $DY \parallel BC$. Prove that $\angle YDA = 2\angle YCA$.

Problem 4. Let $ABCD$ be a convex quadrilateral with $\angle DAB = \angle BCD = 90^\circ$ and $\angle ABC > \angle CDA$. Let Q and R be points on segments BC and CD , respectively, such that line QR intersects lines AB and AD at points P and S , respectively. It is given that $PQ = RS$. Let the midpoint of BD be M and the midpoint of QR be N . Prove that the points M, N, A and C lie on a circle.

Problem 5. Let $ABCD$ be a convex quadrilateral such that the line BD bisects the angle ABC . The circumcircle of triangle ABC intersects the sides AD and CD in the points P and Q , respectively. The line through D and parallel to AC intersects the lines BC and BA at the points R and S , respectively. Prove that the points P, Q, R and S lie on a common circle.

Problem 6. Let ABC be an acute triangle such that $\angle BAC > 45^\circ$ with circumcentre O . A point P is chosen inside triangle ABC such that A, P, O, B are concyclic and the line BP is perpendicular to the line CP . A point Q lies on the segment BP such that the line AQ is parallel to the line PO . Prove that $\angle QCB = \angle PCO$.

Problem 7. Let ABC be a triangle such that $\angle C = 2\angle B$ and ω be its circumcircle. A tangent from A to ω intersects BC at E . Ω is a circle passing through B that is tangent to AC at C . Let Ω intersect AB again at F . K is a point on Ω such that EK is tangent to Ω (where A and K are on different sides of BC). Let M be the midpoint of arc BC of ω (not containing A). Prove that $AFMK$ is a cyclic quadrilateral.

Problem 8. Let ABC be a triangle with incentre I , and let D be a point on line BC satisfying $\angle AID = 90^\circ$. Let the excircle of triangle ABC opposite the vertex A be tangent to BC at A_1 . Define points B_1 on CA and C_1 on AB analogously, using the excircles opposite B and C , respectively. Prove that if quadrilateral $AB_1A_1C_1$ is cyclic, then AD is tangent to the circumcircle of $\triangle DB_1C_1$.